

Intelligent Care Continuity Platform (ICCP):

A Multi-Agent, Explainable-AI-Augmented Framework for Unified Patient-Centered Healthcare Coordination

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Abstract—Healthcare has undergone extensive digital transformation through Electronic Health Records (EHRs), Hospital and Laboratory Information Systems, patient portals, pharmacy platforms, insurance systems, and telemedicine services, yet care delivery remains highly fragmented: no single stakeholder holds a complete view of a patient's journey. This paper presents the Intelligent Care Continuity Platform (ICCP), a two-layer coordination system comprising (i) a non-AI healthcare backbone — a patient health passport, a unified longitudinal timeline, a care-transition manager, medication management, appointment coordination, and an insurance workspace — and (ii) an explainable, multi-agent AI decision-support layer that interprets medical reports, detects care gaps, tracks longitudinal health trends, prepares patients for clinical visits, and explains insurance coverage, all under continuous clinician oversight. ICCP is realized as a single-platform architecture: a Next.js professional web environment serving six role-based portals, a Capacitor-wrapped Next.js patient application for Android, a Spring Boot (Java 21) backend enforcing JWT-based role-based access control, a PostgreSQL data layer, and a Python/FastAPI AI engine orchestrating six specialised agents. We present the problem motivation, system and data architecture, AI methodology, a phased eight-stage development roadmap, and a comparative analysis against existing single-purpose healthcare applications, showing how workflow orchestration, longitudinal intelligence, and explainability jointly address continuity gaps that isolated digitization has not resolved.

Keywords—care coordination, electronic health records, explainable artificial intelligence, multi-agent systems, clinical decision support, health informatics, interoperability, patient engagement

I. INTRODUCTION

Modern healthcare has undergone significant digital transformation through Electronic Health Records (EHRs), Hospital Information Systems (HIS), Laboratory Information Systems (LIS), patient portals, pharmacy systems, insurance platforms, and telemedicine services [1]. Despite this digitization, healthcare delivery remains highly fragmented. Patients still carry printed prescriptions, laboratory reports, discharge summaries, insurance documents, imaging reports, and medication histories across multiple providers, and each stakeholder — hospital, laboratory, pharmacy, insurer, and the patient — maintains only a partial view of the overall care journey.

This fragmentation is not primarily a technology-adoption problem. Landmark studies on hospital-to-primary-care

handoffs report that deficits in communication and information transfer between providers are directly implicated in adverse events, duplicated investigations, and delayed treatment [2], and subsequent work on transitional care shows that patients moving between care settings are especially vulnerable to breakdowns in coordination even when each individual system is well digitized [3]. In other words, the isolated digitization of individual services has not, by itself, produced continuity of care.

Recent reviews of clinical artificial intelligence (AI) reach a complementary conclusion: AI is most effective when embedded into coordinated clinical workflows rather than operating as a standalone chatbot or a single-document interpreter [7], [8]. This motivates a system that treats AI as a decision-support layer riding on top of a robust, non-AI care-coordination backbone, rather than as the product itself.

This paper proposes the Intelligent Care Continuity Platform (ICCP), which combines (a) an interoperable care-coordination backbone spanning patient records, scheduling, medication, and insurance, with (b) an explainable, multi-agent AI layer that augments — rather than replaces — clinical judgment. The main contributions of this paper are:

- A two-layer reference architecture that separates deterministic care-coordination logic from explainable AI decision support, allowing the platform to function as a complete records system even before AI features are enabled.
- A six-agent AI decision-support design (report interpretation, insurance explanation, care-gap detection, health-trend analysis, visit preparation, and an evidence/confidence engine) in which every recommendation is required to carry a supporting guideline reference and a confidence score.
- A single-codebase, multi-surface implementation strategy — Next.js web portals and a Capacitor-wrapped Android patient app sharing one component library and one backend contract — that reduces the duplication typical of parallel web/mobile healthcare products.
- A phased, scope-disciplined development roadmap (Phase 0 through Phase 8) that distinguishes a demonstrable minimum viable product from stretch-goal, production-grade features such as HL7 FHIR interoperability and multi-hospital tenancy.

The remainder of this paper is organized as follows. Section II reviews related work. Section III states the problem and research gap. Section IV describes the proposed ICCP system. Section V presents the system architecture, Section VI the data model, and Section VII the AI methodology. Section VIII covers implementation and technology choices, Section IX the development roadmap, and Section X a

comparative analysis. Section XI discusses expected outcomes and evaluation, Section XII discusses limitations and future work, and Section XIII concludes the paper.

II. RELATED WORK

A. Interoperability and Care Continuity

Mandl and Kohane argue that conventional EHR architectures trap clinical data inside institutional silos and call for platforms built around open, substitutable applications rather than monolithic records systems [1]. The Health Level Seven (HL7) Fast Healthcare Interoperability Resources (FHIR) standard operationalizes this direction by defining resource-based, API-first data exchange across heterogeneous health-IT systems [4]. Separately, the Institute of Medicine's quality-improvement framework identifies "coordinated care" and "timeliness" as two of the six core dimensions of healthcare quality, underscoring that continuity is a first-class design requirement and not an incidental benefit of digitization [5]. Empirical work on hospital discharge and transitional care further shows that communication deficits between care settings are a recurring, measurable source of harm [2], [3], which directly motivates ICCP's Care Transition Manager and Unified Patient Timeline.

B. Artificial Intelligence in Clinical Practice

Large-scale reviews describe a shift from AI as an isolated diagnostic tool toward AI embedded across the clinical workflow, from triage to follow-up [7]. Rajkomar et al. survey machine-learning applications in medicine and emphasize that predictive performance alone is insufficient without integration into clinician workflows and appropriate human oversight [8]. Esteva et al. provide a guide to deep learning in healthcare, cataloguing computer-vision and natural-language tasks — including document and image interpretation — that are directly relevant to ICCP's AI Medical Report Interpreter [9]. Char, Shah, and Magnus highlight the ethical obligations that accompany clinical machine learning, including transparency, accountability, and the avoidance of automation bias, which inform ICCP's requirement that clinicians remain the final decision-makers [10]. Raghupathi and Raghupathi review the broader promise and infrastructure challenges of big-data analytics in healthcare, including the longitudinal, multi-source data integration that ICCP's timeline model is designed to support [6].

C. Multi-Agent Systems and Orchestration

ICCP's AI layer is organized as a set of cooperating, narrowly scoped agents rather than a single monolithic model. This follows the classical multi-agent systems view, in which complex behavior emerges from coordinated, specialized agents rather than one general-purpose reasoner [13]. Applying this principle to healthcare AI keeps each agent auditable and independently testable — a property that is difficult to obtain from a single end-to-end chatbot.

Complementary NLP work on OCR-driven document understanding demonstrates that unstructured or scanned clinical documents — discharge summaries, handwritten prescriptions, laboratory printouts — can be reliably converted into structured fields prior to downstream summarization, which is the specific capability ICCP's AI Medical Report Interpreter builds on rather than reinvents [9].

D. Explainable AI for Clinical Decision Support

Because clinicians and patients must be able to trust AI-generated recommendations, ICCP's Evidence Engine draws on the explainable-AI (XAI) literature. Model-agnostic local explanation techniques such as LIME [11] and SHAP [12] popularized the idea of attaching a human-interpretable justification to any individual prediction; ICCP adapts this principle qualitatively by requiring every AI output to carry a cited guideline and a confidence score rather than a bare recommendation, consistent with the broader informatics literature on clinical decision support [14].

E. Positioning Relative to Commercial Care-Coordination Platforms

Commercial patient-portal and population-health products typically extend a single EHR vendor's data model to additional stakeholders, which improves access within one institution but does little for a patient who moves between unaffiliated hospitals, independent laboratories, community pharmacies, and a separate insurer. Consumer health-record aggregators, at the other extreme, present a merged record but generally stop at read-only aggregation and do not orchestrate workflow (booking, referrals, medication renewal) or attach explanations to what they display. ICCP is positioned deliberately between these two categories: it is vendor-neutral like an aggregator, but it also owns workflow state (Care Transition Manager, Appointment Coordination) like an institutional portal, and it adds a cited, confidence-scored explanation layer that neither category typically provides.

III. PROBLEM STATEMENT AND RESEARCH GAP

A. Background

Modern healthcare generates enormous amounts of patient information throughout diagnosis, treatment, rehabilitation, and long-term monitoring. This information is distributed across disconnected systems owned by hospitals, laboratories, pharmacies, insurance providers, and patients themselves. As patients transition between providers, critical clinical information is frequently unavailable, duplicated, delayed, or misunderstood [2], [3]. Existing healthcare applications generally focus on isolated services — appointment booking, electronic prescriptions, patient portals, laboratory result viewing, or insurance claim management — and rarely provide comprehensive continuity across a patient's entire healthcare lifecycle.

B. Research Gap

Based on the literature reviewed in Section II and an audit of existing commercial and academic healthcare platforms, we identify eight recurring gaps that current systems do not jointly address:

- 1) End-to-end care continuity across independent providers.
- 2) Longitudinal patient intelligence that spans years, not single encounters.
- 3) Cross-platform healthcare coordination between hospitals, labs, pharmacies, and insurers.
- 4) Explainable AI assistance with cited evidence and confidence, not opaque outputs.
- 5) Insurance comprehension support in plain language, ahead of treatment decisions.
- 6) A single, unified clinical timeline rather than per-department records.

7) Workflow automation across stakeholders, not just within one institution.

8) Human-in-the-loop AI governance with clinicians retaining final authority.

Current AI-enabled healthcare products typically interpret individual reports or answer isolated medical questions, but rarely coordinate healthcare activities across multiple providers while preserving explainability and clinician oversight. ICCP is designed to close this specific combination of gaps.

IV. PROPOSED SYSTEM: ICCP OVERVIEW

ICCP introduces a coordination layer connecting patients, healthcare providers, laboratories, pharmacies, and insurers through a unified digital ecosystem. Rather than functioning as another patient portal, the platform continuously maintains a patient's healthcare journey and intelligently assists every stakeholder across the care lifecycle. The architecture consists of two tightly integrated layers, described below.

A. Layer 1 — Core Healthcare Platform

This layer operates independently of AI and is the platform's non-negotiable backbone.

Patient Health Passport — Personal profile, allergies, chronic diseases, surgical history, vaccinations, emergency contacts, medication history.

Unified Patient Timeline — Chronological record of consultations, laboratory reports, radiology, prescriptions, admissions, claims, and follow-ups.

Care Transition Manager — Coordinates movement between primary care, specialists, laboratories, hospitals, rehabilitation, and home care; tracks pending clinical activities.

Medication Management — Schedules, refill reminders, adherence monitoring, and prescription history.

Appointment Coordination — Booking, referral tracking, follow-up scheduling, and reminders.

Insurance Workspace — Policy storage, claim tracking, bills, and coverage information.

B. Layer 2 — AI Decision Support

AI enhances, but never replaces, the clinical workflows defined in Layer 1. Every AI output is advisory and subject to clinician review.

AI Medical Report Interpreter — Converts blood reports, imaging reports, prescriptions, and discharge summaries into patient-friendly explanations via OCR and LLM summarization.

AI Insurance Policy Assistant — Explains coverage, waiting periods, exclusions, and required documents in plain language.

AI Care Gap Detector — Flags missing expected healthcare activities, e.g., an overdue laboratory test for a chronic condition.

AI Health Trend Analyzer — Monitors vitals and laboratory values (glucose, blood pressure, cholesterol, BMI, renal function) across time rather than in isolation.

AI Doctor Visit Preparation — Generates visit summaries, trend reports, and suggested questions ahead of a clinical appointment.

AI Evidence Engine — Attaches a supporting medical guideline and a confidence score to every recommendation produced by the other five agents.

C. Design Principle: Human-in-the-Loop

Consistent with the ethical guidance summarized in Section II-B [10], no AI agent in ICCP is permitted to write directly to the clinical record, alter a prescription, or close a care gap without explicit clinician confirmation. This preserves accountability while still reducing the manual effort of aggregating and interpreting fragmented information. In practice this is enforced structurally rather than by convention: AI agent outputs are persisted to a separate review-queue table with its own access-control rules, and a TimelineEvent is created only after a clinician action moves a queued item to an approved state, so the deterministic Layer 1 data model in Section VI never contains an unreviewed AI claim.

V. SYSTEM ARCHITECTURE

ICCP adopts a single-platform architecture in which every stakeholder interacts with the same backend services while receiving an interface optimized for their responsibilities, rather than maintaining separate web and mobile codebases with duplicated business logic.

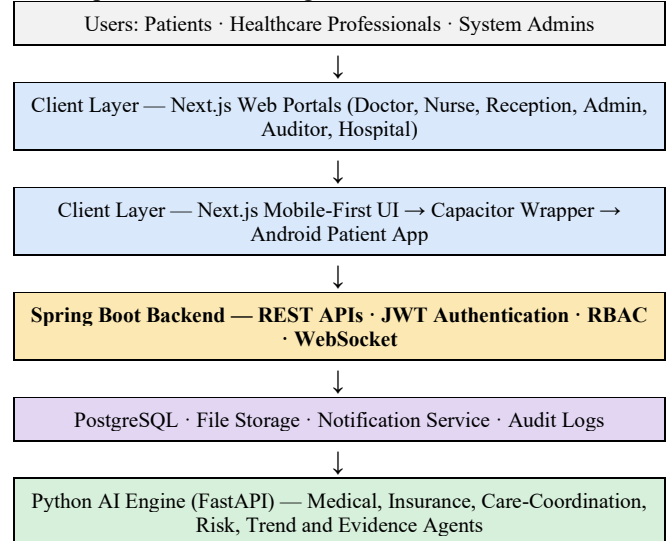


Fig. 1. Unified ICCP system architecture: one backend, role-based professional web portals, and a single Next.js codebase packaged for Android via Capacitor.

Two client environments are supported. The Healthcare Professional Environment is a Next.js web application serving six role-scoped portals (Doctor, Nurse, Reception, Admin, Auditor, Hospital), all sharing one component library while Role-Based Access Control (RBAC) restricts each user to authorized data and actions. The Patient Environment is not a separate codebase: the patient-facing Next.js interface is built mobile-first and packaged into a native Android application using Capacitor, so authentication, API communication, and UI logic remain part of a single project, with Android Studio used only for native packaging, permissions, signing, and publishing.

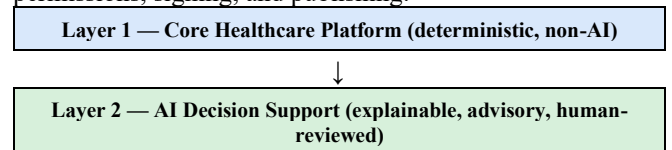


Fig. 2. The two-layer separation between deterministic care coordination (Layer 1) and advisory, explainable AI (Layer 2).

The backend exposes REST APIs from a Spring Boot (Java 21) application, secured with Spring Security and JSON Web Tokens (JWT), and enforces RBAC at the endpoint level rather than only in the client. PostgreSQL is the system of record; Flyway manages schema migrations. The Python/FastAPI AI engine is deployed as an independent service invoked by the backend, so AI features can be scaled, versioned, or temporarily disabled without affecting the deterministic Layer 1 platform.

VI. DATA MODEL

The data model is organized around a central TimelineEvent entity so that every clinical, administrative, and insurance activity can be projected onto one chronological view regardless of which module produced it. Table I summarizes the core entities.

TABLE I CORE DATA ENTITIES

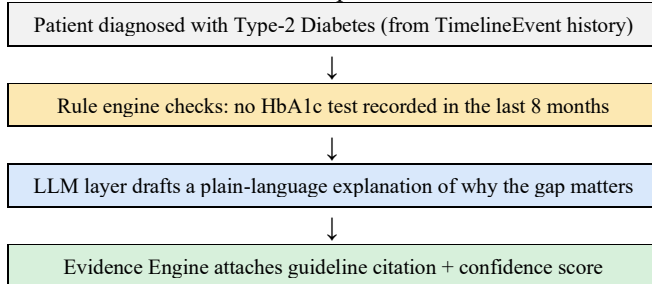
Entity	Key Attributes
User	id, email, password_hash, role, created_at
Patient	id, user_id, dob, gender, blood_group
Provider	id, user_id, specialty, license_number
TimelineEvent	id, patient_id, event_type, payload (jsonb), occurred_at
MedicationSchedule	id, patient_id, drug, dosage, frequency, date_range
Appointment	id, patient_id, provider_id, status, scheduled_at
InsurancePolicy	id, patient_id, provider_name, coverage_details

Storing heterogeneous clinical payloads as JSONB inside TimelineEvent, rather than one table per event subtype, allows new event types (e.g., a new laboratory panel) to be introduced without a schema migration, while a typed event_type enum keeps the timeline queryable and index-friendly.

Two indexing decisions follow directly from this design. First, a composite index on (patient_id, occurred_at) supports the dominant access pattern — rendering a patient's chronological timeline — in a single sorted scan rather than a join across per-module tables. Second, a partial or expression index on event_type within the JSONB payload lets the Care Gap Detector (Section VII) query "most recent event of type X for patient Y" efficiently even as the total number of stored events grows across a patient's lifetime, which is the query this rule-based agent issues most frequently.

VII. METHODOLOGY: MULTI-AGENT AI WORKFLOW

Each Layer 2 capability is implemented as a distinct, narrowly scoped agent rather than a single general-purpose chatbot, following the multi-agent systems principle that specialized, coordinated components are more auditable than one monolithic reasoner [13]. Figure 3 illustrates the reasoning chain used by the AI Care Gap Detector, the agent with the clearest deterministic-plus-AI structure.



System recommends a follow-up consultation for clinician sign-off

Fig. 3. Example reasoning chain for the AI Care Gap Detector agent.

This example is representative of the general pattern used across all six agents: a deterministic rule or retrieval step first narrows the problem using structured TimelineEvent data, after which a language model is used only to summarize, explain, or draft — never to invent clinical facts outside the retrieved context. The AI Evidence Engine is implemented as a shared utility called by the other five agents rather than as a separate agent, ensuring that a citation and confidence score is attached uniformly to every recommendation before it reaches a clinician or patient, consistent with the explainability requirement introduced in Section II-D [11], [12].

The AI Medical Report Interpreter follows OCR extraction (PaddleOCR/Tesseract) with LLM-based structuring and summarization; the AI Insurance Policy Assistant performs retrieval-augmented question answering grounded strictly in the patient's stored policy documents to reduce hallucinated coverage claims; the AI Health Trend Analyzer aggregates time-series vitals and laboratory values before generating a narrative summary; and the AI Doctor Visit Preparation agent composes a pre-visit brief from the timeline and trend outputs. In every case, agent output is written to a review queue, not directly to the clinical record, satisfying the human-in-the-loop principle stated in Section IV-C.

Algorithm 1 gives the deterministic rule that gates the LLM step of the AI Care Gap Detector, illustrating the "rule-first, LLM-explains" pattern used across all six agents so that clinically material decisions never depend solely on unconstrained language-model generation.

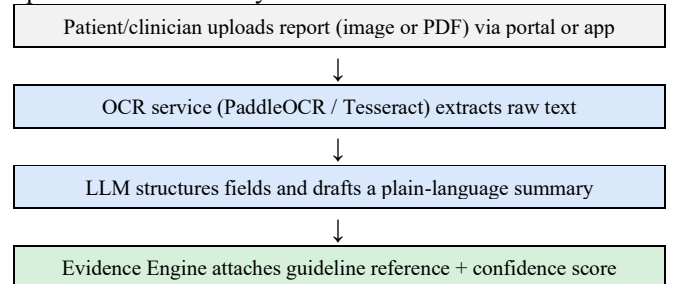
Algorithm 1: Care Gap Detection (per chronic-condition rule)

```

Input: patient_id, condition_code,
       guideline_interval
Output: care_gap (bool), draft_explanation,
        confidence
1: events ← QUERY TimelineEvent WHERE
   patient_id, event_type = test_code
2: last ← MAX(events.occurred_at)
3: if NOW() - last > guideline_interval then
4:   care_gap ← TRUE
5:   draft_explanation ←
     LLM.explain(condition_code, last)
6:   confidence ←
     EvidenceEngine.score(condition_code)
7:   ENQUEUE clinician_review(care_gap,
     draft_explanation, confidence)
8: end if
  
```

Fig. —. Pseudocode for the Care Gap Detector's deterministic gating rule; the LLM is invoked only after the rule fires, and its output is queued for clinician review rather than written directly to the record.

Figure 4 traces the same pattern for the AI Medical Report Interpreter, the agent with the most user-visible latency, from upload to timeline entry.



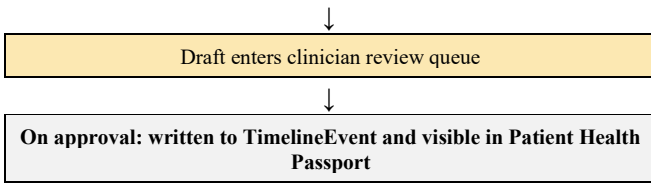


Fig. 4. Request flow for the AI Medical Report Interpreter, from upload to a clinician-approved timeline entry.

VIII. IMPLEMENTATION AND TECHNOLOGY STACK

Table II summarizes the technology stack selected for each architectural layer, chosen to balance developer productivity, type safety on the backend, and access to a mature ML/LLM ecosystem on the AI side.

TABLE II TECHNOLOGY STACK

Layer	Technology
Web Portals	Next.js
Patient Mobile App	Next.js + Capacitor + Android Studio
Backend	Java 21, Spring Boot, Spring Security
Database	PostgreSQL, Flyway migrations, Spring Data JPA
AI Services	Python 3.10+, FastAPI, LangChain / Google ADK
ML / LLM	TensorFlow, PyTorch, Scikit-learn, Grok / Gemini
OCR	PaddleOCR / Tesseract
Notifications	Firebase Cloud Messaging
DevOps	Docker, Docker Compose; AWS S3 and cloud deploy (future)

Authentication and JSON handling on the backend rely on jjwt for JWT issuance and parsing, springdoc-openapi for interactive API documentation, and MapStruct/Lombok to reduce boilerplate in DTO–entity mapping. Real-time features (appointment and lab-result alerts) use Spring WebSocket, and outbound calls to the LLM provider use Spring WebFlux's reactive WebClient to avoid blocking request threads during report interpretation.

A. Representative API Contract

As an illustration of the API contract shared by every client, a report-upload call such as `POST /api/v1/patients/{id}/reports` accepts a multipart file and returns a 202 Accepted response referencing an asynchronous job id, since OCR and LLM structuring (Section VII, Fig. 4) may take several seconds; clients then poll `GET /api/v1/reports/{jobId}` or receive a WebSocket push once the draft has been queued for clinician review. Keeping report interpretation asynchronous, rather than a blocking call, was a deliberate choice so the Android patient app remains responsive on constrained mobile networks.

B. Deployment Topology

For local development and academic demonstration, all services (Spring Boot backend, PostgreSQL, Python AI engine) are orchestrated with Docker Compose on a single host, which keeps the prerequisite list in Section VIII short and reproducible. For a pilot deployment, the same container images are intended to move to separate services behind an API gateway, with PostgreSQL and file storage promoted to managed cloud equivalents (Section VIII, Table II) once the platform moves beyond a single-institution demonstration; this staged deployment path mirrors the phase discipline of Section IX and avoids taking on cloud-operations complexity before the core data model has been validated.

IX. SECURITY, PRIVACY, AND ACCESS CONTROL

Because ICCP centralizes clinical, administrative, and insurance data for a patient across institutions, its security model is treated as a first-class architectural concern rather than a Phase 6 addendum, consistent with the ethical-deployment guidance in Section II-B [10].

A. Authentication and Session Management

All clients authenticate against the Spring Security filter chain and receive a short-lived JWT access token plus a longer-lived refresh token. Tokens carry the user's role and, for provider-side roles, an institution identifier, both of which are validated on every request rather than trusted from the client. Refresh tokens are rotated on use and revocable, so a compromised device can be logged out remotely.

B. Role-Based Access Control

Table V summarizes the coarse-grained permission matrix enforced at the API layer. Enforcement happens in backend middleware/annotations, not only in portal UI logic, so a Nurse-role token cannot reach a Doctor-only endpoint even if it is called directly.

TABLE V ROLE–PERMISSION MATRIX (COARSE-GRAINED)

Role	Representative Permissions
Patient	View own passport/timeline; upload reports; message provider
Doctor	View assigned patients' timeline; write prescriptions; approve AI drafts
Nurse	Record vitals; log medication administration
Reception	Register patients; schedule appointments
Admin	Manage users and roles within an institution
Auditor	Read-only access to audit logs across the institution

C. Data Protection and Auditability

Data is encrypted in transit (TLS) and at rest at the database layer; secrets and API keys (including the LLM provider key) are kept out of source control and injected via environment configuration. Every read or write to a patient's TimelineEvent is written to an append-only audit log, visible to the Auditor role, to support the accountability requirement that motivates Layer 2's human-in-the-loop design (Section IV-C). For an academic prototype, seed and demonstration data are synthetic; no real protected health information is used prior to a formally approved pilot.

X. DEVELOPMENT METHODOLOGY AND ROADMAP

Development is organized into nine phases (Phase 0–8). Phases 0–4 constitute a credible, demonstrable minimum viable product; Phases 5–7 add the AI decision-support layer and production hardening; Phase 8 is explicitly out of scope for the current timeline and reserved for future work. Table III summarizes each phase.

TABLE III DEVELOPMENT PHASES

Phase	Focus	Key Deliverable
0	Foundation & planning	ERD, API contract, repo scaffold, Docker Compose
1	Backend & auth	JWT issuance, RBAC-enforced endpoints
2	Core healthcare modules	Passport, timeline, medication, appointments, insurance APIs
3	Web portals	Doctor, Reception, Admin portals (priority); Nurse/Auditor/Hospital thinner
4	Patient application	Installable Android APK via Capacitor

Phase	Focus	Key Deliverable
5	AI decision support	2–3 fully integrated agents end-to-end
6	Hardening	WebSockets, audit logging, RBAC edge-case testing
7	Testing & demo	E2E tests, seed data, demo script
8	Stretch (future)	HL7 FHIR, insurer integration, multi-hospital tenancy

This phasing deliberately avoids the common failure mode of building AI features before the underlying records platform is reliable: AI outputs are only as trustworthy as the TimelineEvent data they reason over, so Phase 5 is scheduled only after Phase 2 is stable.

XI. COMPARATIVE ANALYSIS

Table IV compares ICCP against the design pattern of existing single-purpose healthcare solutions along the dimensions identified as gaps in Section III-B.

TABLE IV ICCP VS. EXISTING HEALTHCARE SOLUTIONS

Existing Solutions	Proposed ICCP
Focus on one hospital/provider	Coordinates the full patient journey across stakeholders
Static patient records	Longitudinal healthcare timeline
Appointment management only	Care-continuity orchestration
Report viewing	Explainable AI report interpretation
Insurance stored as PDFs	AI-assisted insurance understanding
Basic reminders	Context-aware care-transition management
Standalone AI chatbot	Multi-agent workflow integrated with clinical processes
AI provides answers	AI supports decisions with cited evidence and clinician oversight
Fragmented modules	Unified patient-intelligence platform

The distinguishing claim of ICCP is therefore not the presence of AI, but the combination of workflow orchestration, longitudinal intelligence, and explainability inside one integrated platform, whereas existing tools typically deliver at most one of these properties in isolation.

XII. EXPECTED OUTCOMES AND EVALUATION

As a proposed platform, ICCP's outcomes are stated here as design goals to be validated in a future pilot deployment rather than as measured results. The platform is expected to (a) improve continuity of care across providers by making the unified timeline the default reference point during any consultation; (b) reduce duplicated diagnostic testing by surfacing recent results across institutions before a new test is ordered; (c) improve medication adherence and follow-up compliance through proactive reminders and gap detection; (d) improve patient understanding of clinical and insurance information through plain-language AI explanations; and (e) reduce administrative burden through coordinated, cross-stakeholder workflows.

A future evaluation plan would measure these goals through the indicators summarized in Table VI. No pilot has yet been conducted; the table lists the instrument to be used and the intended direction of change, not measured values, and the figures should be read as hypotheses the architecture is designed to support rather than as reported results.

TABLE VI PROPOSED EVALUATION METRICS

Goal	Metric / Instrument
Care continuity	Duplicate-test rate at partner clinic, pre/post adoption
Medication adherence	Medication possession ratio (MPR)
Care-gap responsiveness	Median time from gap flagged to closure
Usability	System Usability Scale (SUS), patients and clinicians
AI trustworthiness	Clinician agreement/override rate on AI recommendations
Administrative burden	Staff-reported time-to-complete referral workflow

These instruments are standard in health-informatics evaluation and are chosen so that a future pilot can report ICCP's effect using measures comparable to prior care-coordination and clinical-AI studies [2], [3], [8], rather than platform-specific metrics that would be difficult to benchmark against the literature reviewed in Section II.

A staged rollout is recommended for evaluation rather than a single big-bang deployment: an initial silent-mode period in which AI agents populate the review queue without being shown to patients would let clinicians calibrate trust against Table VI's agreement/override metric before patient-facing explanations are enabled, reducing the risk that a low-confidence early recommendation undermines adoption of the platform as a whole.

Silent mode also gives an institution a low-risk way to collect the construct-validity baseline discussed in Section XII-A before any patient-facing claim is made: because AI drafts are logged to the review queue regardless of whether they are shown to patients, the agreement/override rate can be computed from day one of a pilot, well before there is enough longitudinal TimelineEvent history for the Care Gap Detector's recall to be representative of steady-state operation.

XIII. ILLUSTRATIVE CASE WALKTHROUGH

To make the interaction between Layer 1 and Layer 2 concrete, this section traces a single representative patient journey end-to-end through ICCP, referencing the modules described in Sections IV–VII.

A. Scenario

Consider a patient with a prior diagnosis of Type-2 diabetes who visits a new specialist after relocating to a different city. Under the fragmented status quo described in Section I, this specialist would typically start from a blank record: the patient's prior HbA1c history, current medications, and insurance coverage would need to be re-collected verbally or via printed documents, with the accompanying risk of omission described by Kripalani et al. [2].

B. Walkthrough

1. Registration: Reception Portal creates an Appointment linked to the patient's existing Patient Health Passport (Section VI), so the specialist starts from the same TimelineEvent history the patient's previous providers used, not a blank form.

2. Pre-visit preparation: the AI Doctor Visit Preparation agent (Section IV-B) compiles a one-page brief from the Unified Patient Timeline — last HbA1c value and date, current MedicationSchedule, and any open care gaps — and queues it for the specialist to review before the appointment.

3. Care-gap flag: the AI Care Gap Detector (Algorithm 1, Section VII) has already found that the patient's last HbA1c

test is 9 months old against a 6-month guideline interval, and has queued a draft recommendation with an Evidence Engine citation and confidence score.

4. Clinician review: the specialist opens the Doctor Portal, sees the queued brief and the flagged gap side by side, and — using clinical judgment the platform does not override — orders a new HbA1c test and adjusts the medication schedule.

5. Insurance check: before ordering an additional lipid panel, the patient asks about coverage; the AI Insurance Policy Assistant answers from the patient's stored InsurancePolicy document rather than requiring a separate call to the insurer.

6. Timeline update: the new order, the adjusted MedicationSchedule, and the specialist's note are written to TimelineEvent, so the next provider the patient sees — in this city or another — starts from the same up-to-date picture.

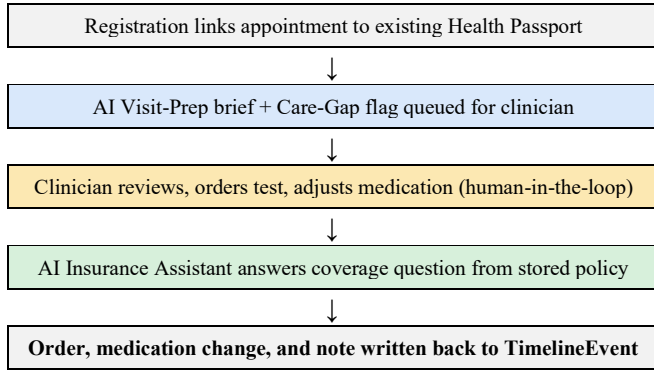


Fig. 5. End-to-end trace of the illustrative case through Layer 1 (deterministic) and Layer 2 (advisory AI) modules.

C. What the Walkthrough Demonstrates

Three properties of the architecture are visible in this single journey. First, continuity: the specialist never re-collects information the platform already has, directly addressing the handoff-communication gap identified in Section III-B. Second, explainability: both the care-gap flag and the insurance answer arrive with a traceable source (a guideline citation and a stored policy document, respectively) rather than as unattributed AI output, consistent with Section II-D. Third, human authority: at every decision point — ordering the test, adjusting medication — the platform surfaces information but the clinician acts, matching the design principle stated in Section IV-C and the ethical guidance in Section II-B [10]. None of the six steps above requires functionality beyond what Sections IV–VIII already specify; the walkthrough is a composition of existing modules, not an additional feature.

XIV. DISCUSSION, LIMITATIONS, AND FUTURE WORK

Several limitations follow directly from ICCP's current scope. First, Phases 0–4 alone constitute a legitimate, demoable records system, but the full six-agent, six-portal, FHIR-ready vision described in Section IV requires the stretch-goal Phase 8 work; any written evaluation of a prototype build should scope its claims to whichever phases were actually implemented, rather than the complete architecture. Second, because this is healthcare data, any deployment — even an academic prototype — must document its handling of personally identifiable and protected health information, including encryption at rest and in transit, JWT expiry/refresh policy, and RBAC edge-case testing (for example, verifying that a Nurse-role token cannot

reach a Doctor-only endpoint). Third, the AI layer's usefulness is bounded by data completeness: care-gap detection and trend analysis are only as reliable as the TimelineEvent history captured in Layer 1, so early deployments with sparse data should present AI outputs with appropriately lower confidence.

Future work includes HL7 FHIR interoperability for exchange with external EHR systems [4], real integrations with insurer, laboratory, and pharmacy systems in place of the current stubbed claim tracking, a dedicated Risk Prediction Agent once sufficient longitudinal data is available, and multi-hospital tenancy to support health-system-scale deployment. Clinical validation of AI Care Gap Detector recommendations against real clinician judgment, ideally in a prospective pilot, is a prerequisite before any production deployment.

Finally, the roadmap in Section IX is written to remain honest about what has actually been built at any given time: because Phases 0–4 are independently demoable, a report or paper describing an in-progress ICCP implementation should state precisely which phases were completed rather than describing the full eight-phase vision as delivered. This scope discipline is treated as a design requirement of the project's own documentation practice, not only of the software.

A. Threats to Validity

Three threats to validity are worth stating explicitly for future evaluators of this architecture. First, construct validity: the evaluation metrics proposed in Table VI (Section XI) are proxies — for example, medication possession ratio approximates but does not directly measure adherence — and should be interpreted alongside qualitative clinician feedback rather than in isolation. Second, external validity: a pilot conducted within one partner clinic or teaching hospital may not generalize to settings with different staffing ratios, patient literacy levels, or connectivity constraints, which is why Section IX intentionally scopes early phases to a single-institution demonstrable MVP before any multi-hospital claim is made. Third, measurement validity of the AI layer specifically: because the Care Gap Detector's rule-first design (Algorithm 1, Section VII) depends on complete TimelineEvent history, a clinic that has not yet migrated its historical records will see artificially low gap-detection recall until back-filling is complete — a limitation that should be disclosed in any pilot report rather than attributed to the underlying method.

B. Broader Ethical and Adoption Considerations

Beyond the technical threats to validity above, two adoption-facing considerations bear on whether ICCP's design goals are achievable in practice. Algorithmic-bias risk is inherited from whichever LLM provider and OCR engine are configured (Section VIII, Table II): report interpretation and insurance explanations should be periodically audited for systematic errors across patient subgroups, not only for aggregate accuracy, before broader rollout. Clinician workload is also a genuine risk in the opposite direction from what AI assistance intends: a review queue that surfaces too many low-confidence drafts can create more work than it saves, so the confidence threshold used to enqueue an item for clinician review (Algorithm 1, Section VII) should be tunable per institution and revisited using the agreement/override metric in Table VI rather than fixed at build time.

C. Cost and Operational Considerations

Two operational costs are worth flagging alongside the technical roadmap in Section IX. LLM inference cost scales with the volume of report-interpretation and insurance-assistant calls (Section VII), so a production deployment should budget for response caching on repeated or near-duplicate queries — for example, re-explaining the same standard insurance exclusion clause across many patients — rather than treating every request as a fresh LLM call. Second, OCR accuracy on handwritten prescriptions and low-quality scans remains materially lower than on typed lab reports; institutions piloting ICCP should expect the AI Medical Report Interpreter's confidence scores to be visibly lower on handwritten input, and clinician review time should be budgeted accordingly rather than assumed away by the presence of OCR.

D. Summary of AI Agent Responsibilities

Table VII consolidates the six Layer 2 agents introduced across Sections IV and VII into a single reference, alongside the primary Layer 1 data each agent consumes and the human checkpoint it is gated behind, to make the human-in-the-loop boundary auditable at a glance.

TABLE VII AI AGENT SUMMARY AND HUMAN CHECKPOINTS

Agent	Input Consumed	Human Checkpoint
Report Interpreter	Uploaded document (image/PDF)	Clinician approves before TimelineEvent write
Insurance Assistant	Stored InsurancePolicy	Patient/clinician reads; no auto-action
Care Gap Detector	TimelineEvent history	Clinician confirms before gap closes
Health Trend Analyzer	Vitals/labs time-series	Clinician interprets trend narrative
Visit Preparation	Timeline + trend output	Clinician reviews brief pre-visit
Evidence Engine	Any of the above agents' draft	Shared utility; no independent action

Presenting the agents this way makes explicit that Section IV-C's human-in-the-loop principle is not a single global switch but six independent checkpoints, one per agent, each of which must be satisfied before that agent's output can affect the patient record.

XV. CONCLUSION

This paper presented the Intelligent Care Continuity Platform (ICCP), a two-layer system that pairs a deterministic care-coordination backbone with an explainable, multi-agent AI decision-support layer under continuous clinician oversight. By separating longitudinal record-keeping and workflow orchestration (Layer 1) from advisory AI reasoning (Layer 2), and by requiring every AI recommendation to carry a cited guideline and confidence score, ICCP targets the specific combination of continuity, explainability, and cross-stakeholder coordination that recent literature and our comparative analysis suggest existing single-purpose healthcare applications do not jointly provide. A phased, scope-disciplined roadmap distinguishes an achievable minimum viable product from stretch-goal production features, and a concrete evaluation plan is proposed for future pilot validation. We regard ICCP as a reference architecture rather than a finished product, and we hope it is useful to other groups designing care-coordination systems that must earn clinician trust before earning clinician adoption.

The illustrative case walkthrough in Section XI shows that this trust is earned incrementally, one reviewed recommendation at a time, rather than granted by the presence of an AI layer on its own; the architecture's central bet is that a platform which is honest about what it does and does not know — through Table VI's metrics, Algorithm 1's confidence gating, and the Evidence Engine's citations — is more likely to be adopted by clinicians than one that presents AI output as authoritative by default.

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REFERENCES

- [1] K. D. Mandl and I. S. Kohane, "Escaping the EHR trap — the future of health IT," *N. Engl. J. Med.*, vol. 366, no. 24, pp. 2240–2242, 2012.
- [2] S. Kripalani et al., "Deficits in communication and information transfer between hospital-based and primary care physicians: implications for patient safety and continuity of care," *JAMA*, vol. 297, no. 8, pp. 831–841, 2007.
- [3] E. A. Coleman and R. A. Berenson, "Lost in transition: challenges and opportunities for improving the quality of transitional care," *Ann. Intern. Med.*, vol. 141, no. 7, pp. 533–536, 2004.
- [4] HL7 International, "HL7 FHIR Release 4 (R4)," *Fast Healthcare Interoperability Resources specification*, 2019.
- [5] Institute of Medicine (Committee on Quality of Health Care in America), *Crossing the Quality Chasm: A New Health System for the 21st Century*. Washington, DC: National Academies Press, 2001.
- [6] W. Raghupathi and V. Raghupathi, "Big data analytics in healthcare: promise and potential," *Health Inf. Sci. Syst.*, vol. 2, no. 1, art. 3, 2014.
- [7] E. J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nat. Med.*, vol. 25, pp. 44–56, 2019.
- [8] A. Rajkomar, J. Dean, and I. Kohane, "Machine learning in medicine," *N. Engl. J. Med.*, vol. 380, no. 14, pp. 1347–1358, 2019.
- [9] A. Esteva et al., "A guide to deep learning in healthcare," *Nat. Med.*, vol. 25, pp. 24–29, 2019.
- [10] D. S. Char, N. H. Shah, and D. Magnus, "Implementing machine learning in health care — addressing ethical challenges," *N. Engl. J. Med.*, vol. 378, no. 11, pp. 981–983, 2018.
- [11] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?: Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144.
- [12] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. 31st Int. Conf. Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4768–4777.
- [13] M. Wooldridge, *An Introduction to MultiAgent Systems*, 2nd ed. Chichester, U.K.: Wiley, 2009.
- [14] E. Coiera, *Guide to Health Informatics*, 3rd ed. Boca Raton, FL: CRC Press, 2015.